

ECEC state demand inputs: method

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The simulator needs one small file per geography: six rows, one per care type, one number each, the annual hours of child care children under five use. This memo says where that number comes from.

Everything happens at the state level. County data enters once, in the population step, and it enters as a sum. Nothing joins on county.

1 Why this is not a survey lookup

The simulator calibrates each state against the mix of child care its families actually use. NSECE, the survey that measures that mix, does not record which state a family is in. It publishes Census region and nothing finer.

So I rebuild the mix from each state's own administrative records, and I use the survey only for the things no state records can see.

2 Results

The build measures two things. Everything else is a subdivision of them.

	Center-based	Not center-based	Total	Share center
Kentucky	81,326	80,604	161,930	50%
New Jersey	173,225	136,721	309,946	56%
New York	270,982	378,847	649,829	42%
New York City	132,065	169,911	301,976	44%
Virginia	135,068	127,932	263,000	51%

Care that is not center-based is one part counted and one part left over:

	Licensed home care	Informal, unrecorded
Kentucky	4,756	75,848
New Jersey	1,987	134,734
New York	96,366	282,481
New York City	38,642	131,269
Virginia	4,743	123,189

New York's licensed home care is twenty times Kentucky's or Virginia's. That is real rather than an artefact: New York registers home care from three children up, so care that leaves no trace elsewhere sits inside its register.

Children, by care type. These are counts of children, not records and not weighted survey figures:

	KY	NJ	NY	NYC	VA
Center-Based	37,550	116,402	168,235	86,337	87,192
Unpaid Center-Based	43,776	56,823	102,747	45,728	47,876
Paid Home-Based	4,756	1,987	96,366	38,642	4,743
Other Unpaid	45,830	81,411	170,686	79,318	74,436
Other Paid	27,363	48,607	101,909	47,357	44,442
Unpaid Home-Based	2,655	4,715	9,886	4,594	4,311
Total	161,930	309,946	649,829	301,976	263,000

Annual hours, in millions. This is the output file:

	KY	NJ	NY	NYC	VA
Center-Based	63.3	196.2	283.6	145.5	147.0
Unpaid Center-Based	62.9	81.6	147.6	65.7	68.8
Paid Home-Based	9.1	3.8	185.0	74.2	9.1
Other Unpaid	88.4	157.1	329.3	153.0	143.6
Other Paid	42.5	75.5	158.4	73.6	69.1
Unpaid Home-Based	5.6	10.0	21.0	9.7	9.1

New York City is both inside New York and a build of its own. The two overlap deliberately, and they do not match: each is calibrated against its own non-parental total, so the five boroughs come out differently depending on which build produced them. Read one or the other, not both.

3 The build

1. Count the children

Census population estimates, vintage 2019, children aged 0 to 4, summed from counties to the state.

Children 0–4, 2019	
Kentucky	272,610
New Jersey	514,690
New York	1,127,001
New York City	523,718
Virginia	505,477

I use ages 0 to 4 because the simulator does. Its ACS processing filters `age < 5` in seven places, and the demand file has to describe the same children the simulator is calibrating against. A 0–5 universe was built and discarded: it raises every total by 15 to 23 percent, it needs a second population source, and Census publishes no single-year-of-age county estimate, so New York City could only be estimated rather than counted.

New York City is 47 percent of the state’s young children.

2. How many are in someone else’s care

I multiply by a rate from a household survey. NSCH 2019 asks whether the child was in any non-parental care in the past week, and it records the state.

	Children 0–4	Rate	In non-parental care	Records
Kentucky	272,610	59.4%	161,930	147
New Jersey	514,690	60.2%	309,946	101
New York	1,127,001	57.7%	649,829	112
New York City	523,718	57.7%	301,976	112
Virginia	505,477	52.0%	263,000	121

That product is the control total. Every later count is a share of it.

The total is always 2019, because both inputs are 2019. That is what lets me use a provider register published in 2026 without ageing it backward: the register decides the mix, never the level.

New York City takes the state’s rate, because NSCH publishes no sub-state geography. Everything that distinguishes the city has to come from its registers, and it does.

This rate is the softest number in the build. About 120 families per state, which is roughly a seven point standard error on a figure that multiplies the entire non-parental total. Section 5 says why I did not pool more waves to fix it.

3. Turn provider registers into children

Every state licenses child care providers and publishes the register. The register gives capacity; I need children. Three things have to happen, and each state needs them done differently because the registers differ.

Remove the school-age capacity. Every register mixes it in.

- **Kentucky** does not say which providers are centers and which are homes. I split on the state’s own rule: a licensed home is capped at 12 children, so capacity of 12 or less is a home. That gives 1,609 centers and 335 homes. Providers serving school-age children only are dropped. Mixed age bands scale by ratios measured inside the file itself, 0.487 to 0.827 depending on the band.
- **New Jersey** publishes an age range per center. I drop the 1,035 centers serving only ages 6 to 13, and scale mixed ranges by 0.820.
- **New York** is the clean case. It publishes capacity by age, so infant plus toddler plus preschool is the under-five figure exactly. Homes report one lump, but school-age capacity is published separately and subtracts.
- **Virginia** writes the age range as text. I parse it to months and drop anything starting at age five or later. Centers reaching past six scale by 0.60.

Bring the register back to 2019. Two of them are 2026 exports.

- **New York** scales by provider type against the state’s own 2015–2025 capacity series. Family homes have closed in large numbers: group family day care scales to 0.86 of its current size statewide and 0.84 in the city. Centers barely moved, so one blanket factor would distort the mix.

- **Virginia** scales by one statewide factor, 0.90, against its June 2020 capacity matrix.
- **Kentucky and New Jersey** need nothing. Kentucky’s register is a September 2018 snapshot, ten months before the base year and the best vintage here. New Jersey’s is frozen at May 2019.

Turn capacity into children. Capacity is the legal maximum. Real enrollment is lower.

- **Kentucky measured its own, and uses no fill rate at all.** Its 2019 workforce study prints capacity and enrollment side by side: 42 children per center, 9 per home. This is the best way to do it, and it is why 1,609 centers yield only 67,578 children.
- **New Jersey measured its own too,** 67 percent full, from its 2021 market rate survey.
- **New York and Virginia have published none,** so they borrow the national figures: centers 90 percent full, homes 60 percent. New York’s homes take 80 percent instead, the fill rate without the age correction, because its register already removed school-age capacity and applying the correction again would remove it twice.

		Providers	Under-5 capacity	Children
Kentucky	Centers	1,609	107,369	67,578
Kentucky	Homes	335	2,115	3,015
New Jersey	Centers	3,128	251,159	169,106
New York	Centers, state	2,073	153,053	138,191
New York	Centers, city	2,310	141,300	127,580
New York	Homes	11,918	117,401	93,381
New York City	Centers	2,310	141,300	127,580
New York City	Homes	4,580	46,057	36,634
Virginia	Centers	3,186	207,835	187,654
Virginia	Homes	1,935	12,633	7,604

New York needs two registers, because it has two regulators. The state licenses no centers inside New York City; the city permits its own under Health Code Article 47. The state’s row for the city is zero, and that is correct rather than a join failure. Neither register alone covers the state.

4. Add the programs that count their own children

Head Start, Early Head Start and public pre-K publish actual enrollment, so no estimate is needed.

	Pre-K	Head Start	Early Head Start	Total
Kentucky	28,465	12,307	3,004	43,776
New Jersey	41,305	11,940	3,578	56,823
New York	54,451	36,863	11,433	102,747
New York City	26,717	13,073	5,938	45,728
Virginia	33,790	11,579	2,507	47,876

These children come out of the licensed center count, not on top of it. A Head Start center usually holds a state licence, so its children already sit inside step 3’s capacity, and adding them would count them twice.

A published count is treated as a census and does not move in step 5.

This category is an upper bound in every state. I cannot measure how much Head Start overlaps state pre-K, and the published figures do not net it out.

5. Scale the level to the survey

Provider registers are good at shape and bad at level. They say which care types are common in a state; they overstate how many children are in formal care.

So I scale each geography by one factor until formal care matches the share the survey finds, a little over half of all non-parental care. The program counts hold still. Everything else moves together, so the mix is untouched and only the level changes.

	Scaled by	Formal	Informal	Informal share
Kentucky	1.58	86,082	75,848	47%
New Jersey	1.04	175,212	134,734	44%
New York	1.03	367,348	282,481	44%
New York City	1.05	170,707	131,269	44%
Virginia	0.62	139,811	123,189	47%

This table is the best diagnostic in the build.

New Jersey, New York and the city barely move. Their raw register counts landed within two points of the survey before anything was done to them. Three separate state registers agreeing with an independent national survey is the strongest evidence here that the conversion in step 3 is sound.

Virginia is cut by a third. Its register is the most inclusive of the four, listing unlicensed and voluntarily registered family homes that other states never record, and it borrows the national fill rate.

Kentucky is raised by half, and that is the informative one. Kentucky is the only state that measured how full its own centers are: 57 percent, against the 90 percent that New York and Virginia borrow. The borrowed figure is too high by more than half. Kentucky moving up and Virginia moving down is exactly what that gap predicts, so the scaling is correcting a bias two states have measured rather than absorbing an unexplained one.

6. Finish the rows

There is no price split. An earlier version divided center care into low-priced and high-priced. Nothing in the state data supports that division. A state publishes one price, so every state landed wholly on one side of the national line, and a care type with no children in it reads as unmeasured rather than as absent. Center care is now one row, at 1,686 hours per child per year, the national average across both price bands.

Fill in the care nobody records. Three of the six types appear in no register anywhere: unpaid relatives, nannies, informal arrangements. Unpaid relative care is not a business and files no return. It is whatever is left of the control total after everything countable is subtracted, divided three ways on the national pattern. That remainder runs 44 to 47 percent of non-parental care here, against 47 percent nationally.

Convert children to hours. Each care type has a measured hours figure from the national survey: the child-weighted mean weekly hours in the child's primary arrangement, times 50 weeks. They run from about 1,440 hours a year for program center care to about 2,120 for unpaid home care, so the hours column is not proportional to the children column.

4 What each number rests on

	Where it comes from	How solid
Children 0–4	Census, 2019	Counted
Share in non-parental care	Household survey, 2019, about 120 families per state	Soft. Pooling was tested and rejected
Provider capacity	State licensing register	Counted, but a legal maximum. Virginia providers say they can serve 27 percent fewer children than they are licensed for
How full providers are, KY and NJ	The state’s own survey	Measured in state
How full providers are, NY and VA	National survey	Borrowed. Two states’ own data say it is too high
Program enrollment	Federal and state reporting	Counted
Informal split, hours per child	National survey	Borrowed

The non-parental share stays a single 2019 wave

Pooling survey waves would cut the sampling error by 40 to 47 percent. I tested it and rejected it, on the standard that every input matches the simulator’s 2019 base year.

Pooling 2018, 2019 and 2020 centers exactly on 2019, which is the one pool that passes the vintage test. It fails on content. NSCH 2020 was fielded through the pandemic and child care use collapsed:

	2018	2019	2020
Kentucky	57.3%	59.4%	45.8%
New Jersey	58.3%	60.2%	42.6%
New York	64.7%	57.7%	62.1%
Virginia	61.7%	52.0%	46.7%

That is a real shock, not sampling noise, and pooling it in would push every state’s level down by several points. Pooling 2018 and 2019 alone centers on mid-2018 while every other input is pinned to 2019.

So the rate keeps its full sampling error. Both extra waves are held in the repository so the rejection is reproducible, not so the pool can be used.

The borrowed fill rates are too high, measurably

Two states have published their own figures and both point the same way.

Kentucky measured its centers at 57 percent full in 2019, against the national 90 percent that New York and Virginia borrow.

Virginia measured something related in 2022: what providers say they can actually serve, against what they are licensed for. Centers came in at 73 percent of authorized capacity and homes at 91 percent, across 1,968 providers. In that sample, 121,260 licensed slots against 89,374 children providers said they could take.

Neither figure is used as a rate. Kentucky's applies to Kentucky. Virginia's is a 2022 survey and measures capacity rather than enrollment, so it would double-count against the fill rate. Both are evidence about direction and size, and they agree: licensed capacity overstates real supply by roughly a quarter to a third, and the national fill rate does not remove enough of it.

5 What this method does not do

1. **It cannot see unpaid relative care**, the largest informal category. That number is a remainder, not a measurement, and no administrative source anywhere would fix it. Only a restricted-access survey geocode file would.
2. **The mix carries the register's date**. The level is 2019 by construction, but how it divides across care types reflects whenever the state last published. Composition genuinely drifts, and a 2026 register understates family homes for 2019.
3. **Unpaid Center-Based is an upper bound** in every state, because Head Start and state pre-K overlap by an amount nobody publishes, and a program site may also hold a state licence.
4. **The scaling in step 5 forces an exact match** to the survey. That is a choice rather than a finding, and it is worth revisiting: the alternative is to let the registers say what they say and report the disagreement.
5. **The simulator still expects seven rows**, with center care split by price. It rejects a file with six. Bridging the two is a mechanical step, dividing the center row on a national share, and it is deliberately outside this method so that the build does not appear to measure something it cannot.